Optimal Market Making

MATH 70125 Market Microstructure Report

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Abstract

Guéant (2017) reduces the Hamilton–Jacobi–Bellman equation of the optimal market-making problem to a system of ordinary differential equations indexed by time and inventory alone. This report replicates the core numerical experiments of the paper, validates the optimal strategy by Monte Carlo simulation, and backtests it on real order-book data (LOBSTER equities and Binance Futures). All qualitative patterns are confirmed: inventory-dependent quote adjustment, asymptotic convergence, U-shaped spread, Model A/B near-equivalence, and multi-asset cross-inventory coupling. The Monte Carlo validation shows the main advantage of the optimal strategy is risk control (Sharpe 0.71 vs. 0.43), not mean PnL.

We then extend the framework by replacing Poisson arrivals with self-exciting Hawkes processes. This destroys the dimensional reduction: the reduced value function satisfies a four-dimensional PDE with no ODE solution. A Soft Actor–Critic agent trained on the extended problem outperforms the Poisson-optimal baseline by an amount growing monotonically with self-excitation intensity, reaching +15% in adjusted PnL at $\alpha/\beta = 0.2$, well below the near-unity branching ratios observed empirically, suggesting the advantage would be substantially larger in real markets.

Keywords: *market making, stochastic optimal control, inventory risk, Hawkes process, limit order book, reinforcement learning.*

1 Introduction

1.1 The Market-Making Problem

A market maker is a financial intermediary who provides continuous liquidity by simultaneously posting bid and ask prices. By quoting on both sides of the order book, the market maker earns the bid–ask spread on each round-trip transaction, but assumes the risk of accumulating an undesired inventory when buy and sell orders arrive asymmetrically. The central problem is to determine how far from the mid-price each quote should be placed, given the current inventory, the volatility of the asset, and the characteristics of order flow. This is naturally formulated as a stochastic optimal control problem in which the state evolves through a combination of continuous price dynamics and discrete trade events.

1.2 From Avellaneda-Stoikov to Guéant

Avellaneda and Stoikov [2] introduced the first rigorous stochastic control formulation of the market making problem. In their model the mid-price follows an arithmetic Brownian motion, fills arrive as Poisson processes with intensity $\Lambda(\delta) = Ae^{-k\delta}$, and the market maker maximises the expected CARA utility of terminal wealth. The CARA structure allows a partial simplification: the ansatz $u = -e^{-\gamma x}\,\phi(t, s, q)$ separates the cash variable x from the HJB equation. However, the reduced function ϕ still depends on the mid-price s through the diffusion term $\frac{1}{2}\sigma^2\partial^2_{ss}\phi$. To obtain tractable formulas, Avellaneda and Stoikov perform two additional approximations: they expand ϕ asymptotically in the inventory variable q (treated as continuous), and linearise the order arrival terms. This yields the well-known reservation price $r = s - q\gamma\sigma^2(T - t)$ and optimal spread $\delta^a + \delta^b = \gamma\sigma^2(T - t) + \frac{2}{\gamma}\ln(1 + \gamma/k)$. In these formulas, inventory affects quotes only through a linear shift of the reservation price; the total spread itself does not depend on q (a limitation of the approximation not of the underlying model).

Guéant, Lehalle, and Fernandez-Tapia [5] addressed these limitations by introducing two structural changes: bounded inventory ($q \in \{-Q\Delta, \dots, Q\Delta\}$) and a different form of the CARA ansatz, $u = -\exp(-\gamma(x + qS + \theta(t, q)))$. The key difference is the term qS inside the exponential: when computing $\partial^2_{SS}u$, it produces $\gamma^2 q^2\sigma^2 \cdot u$, which after dividing through depends only on q, not on s. This eliminates the mid-price *exactly*, reducing the four-variable HJB to a finite system of ODEs in (t, q) alone. For exponential intensities, a further change of variables linearises the system into a tridiagonal structure solvable in $O(Q)$ per time step.

Guéant (2017) [1] extends this framework in several directions. First, the ODE reduction is shown to hold for *general* intensity functions satisfying mild regularity conditions, not just the exponential case. Second, a unified parametrisation with $\xi \geq 0$ encompasses both the CARA utility criterion ($\xi = \gamma$, Model A) and a running inventory-penalty criterion ($\xi = 0$, Model B), with rigorous existence, uniqueness, and verification theorems. Third, generalised closed-form approximations are derived that improve upon the original Avellaneda–Stoikov formulas: the exact spread is U-shaped in inventory (not constant), and the exact skew is nonlinear (not affine). Fourth, the framework extends to multiple correlated assets, yielding closed-form approximations for the optimal quotes of a multi-asset market maker. These features make the framework both analytically tractable and practically relevant, and explain its central role in the modern optimal market-making literature.

1.3 Scope of This Report

This report has three main objectives. First, we replicate the core numerical results of the paper. Second, we validate the framework through Monte Carlo simulation and an empirical backtest order-book data. Third, we extend the model by replacing Poisson arrivals with Hawkes processes, showing that this breaks the ODE tractability and motivates a reinforcement learning approach.

We reproduce the paper's qualitative and numerical patterns, but not its exact archival values or CDX calibra-

tion procedure. Unless stated otherwise, experiments are conducted with exponential intensities and zero terminal penalty. The full implementation is available at https://github.com/hamzaelarji/Market-Microstructure-.

2 Summary of the Article

The market-making problem can be formulated as a stochastic optimal control problem, where the optimal quoting strategy is characterised by a Hamilton–Jacobi–Bellman (HJB) equation. In practice, solving this equation requires reducing its dimensionality to obtain a tractable representation.

Throughout this report, we follow a common three-step approach: (i) formulate the HJB equation, (ii) identify a suitable reduction of the state space, and (iii) solve the resulting problem analytically or numerically.

The key difference between the models lies in the second step. Avellaneda and Stoikov rely on approximations to obtain explicit formulas, Guéant achieves an exact reduction to an ODE system, while our Hawkes extension shows that no such reduction is available, requiring a fundamentally different numerical approach.

2.1 Model Setup and State Dynamics

The reference price follows an arithmetic Brownian motion $dS_t = \sigma\, dW_t$ with $\sigma > 0$. The absence of drift reflects the assumption that over short market-making horizons, expected returns are negligible. The market maker posts a bid at $S_t - \delta_t^b$ and an ask at $S_t + \delta_t^a$, where $\delta_t^b, \delta_t^a \geq 0$ are the distances from mid-price. Executions occur at the arrival times of two point processes with intensities

$$\lambda_t^b = \Lambda(\delta_t^b)\, \mathbf{1}_{q_{t-} < Q\Delta}, \qquad \lambda_t^a = \Lambda(\delta_t^a)\, \mathbf{1}_{q_{t-} > -Q\Delta},$$

where each fill changes inventory by $\pm\Delta$ (one lot) and Q is an integer inventory limit. The indicator functions enforce that the market maker stops quoting on a given side when the inventory hits the boundary. The intensity functions Λ^b, Λ^a are assumed to be twice differentiable, strictly decreasing, vanishing at infinity, and satisfying a concavity condition ensuring uniqueness of the Hamiltonian optimiser. For all numerical experiments $\Lambda(\delta) = Ae^{-k\delta}$.

The inventory and cash processes evolve as :

$$\begin{aligned} dq_t &= \Delta\, dN_t^b - \Delta\, dN_t^a, \\ dX_t &= (S_t + \delta_t^a)\Delta\, dN_t^a - (S_t - \delta_t^b)\Delta\, dN_t^b. \end{aligned}$$

These equations show that inventory moves with trade executions, while cash decreases on buys and increases on sells at the quoted prices.

2.2 Two Objective Functions

Model A - CARA utility: The market maker maximises

$$\mathbb{E}[-e^{-\gamma(X_T + q_T S_T - \ell(|q_T|))}]$$

where $\gamma > 0$ is a risk-aversion parameter and ℓ is a terminal inventory penalty. The CARA utility penalises both the level and variance of terminal wealth, making the market maker averse to all sources of risk.

Model B - running inventory penalty: The market maker maximises

$$\mathbb{E}[X_T + q_T S_T - \ell(|q_T|) - \frac{\gamma\sigma^2}{2}\int_0^T q_t^2\, dt]$$

This criterion is risk-neutral with respect to execution uncertainty but adds a quadratic running cost penalising the instantaneous inventory variance. The coefficient $\frac{1}{2}\gamma\sigma^2$ ensures that γ plays the same dimensional role as in Model A. Model B is algebraically simpler and produces near-identical quotes in practice.

2.3 The Central Reduction: HJB to ODE

The four-variable HJB equation in (t, x, q, S) is reduced to a system of ODEs in (t, q) alone by exact ansatz.

For Model A, the substitution $u(t, x, q, S) = -\exp(-\gamma(x + qS + \theta(t, q)))$ eliminates both cash and price.

For Model B, the linear ansatz $u(t, x, q, S) = x + qS + \theta(t, q)$ achieves the same.

Introducing the Hamiltonian

$$\begin{aligned} H_\xi(p) &= \frac{A\Delta}{k}\, C_\xi\, e^{-kp}, \\ C_\xi &= \begin{cases} (1 + \xi\Delta/k)^{-(k/(\xi\Delta)+1)} & \xi > 0, \\ e^{-1} & \xi = 0 \end{cases} \end{aligned}$$

both models reduce to the unified ODE:

$$\begin{aligned} 0 = -\partial_t\theta + \tfrac{1}{2}\gamma\sigma^2 q^2 - \mathbf{1}_{q<Q\Delta} H_\xi\left(\frac{\theta_q - \theta_{q+\Delta}}{\Delta}\right) \\ - \mathbf{1}_{q>-Q\Delta} H_\xi\left(\frac{\theta_q - \theta_{q-\Delta}}{\Delta}\right) \end{aligned}$$

with $\xi = \gamma$ for Model A and $\xi = 0$ for Model B.

The dimensionality drops from four to two: θ depends only on (t, q), with q taking finitely many values. The paper establishes existence, uniqueness, and a verification theorem confirming that the derived controls are optimal.

The substitution $v_q(t) = \exp(k\theta(t, q)/\Delta)$ then transforms the nonlinear ODE into a linear tridiagonal system:

$$\begin{aligned} 0 = -\partial_t v_q + \tfrac{1}{2}k\gamma\sigma^2 q^2\, v_q \\ - AC_\xi(\mathbf{1}_{q<Q} v_{q+\Delta} + \mathbf{1}_{q>-Q} v_{q-\Delta}), \end{aligned}$$

with terminal condition $v_q(T) = \exp(-k\,\ell(|q|)/\Delta)$.

2.4 Closed-Form Approximations and Multi-Asset Extension

By replacing the discrete inventory grid with a continuous domain and studying the long-horizon regime $T \to \infty$, the paper derives tractable closed-form approximations (generalised GLF formulas). For exponential intensities under Model B:

$$\begin{aligned} \delta^{\{b,a\}*}(q) &\approx \frac{1}{k} \pm \frac{2q \pm \Delta}{2}\,\omega, \\ \omega &= \sqrt{\frac{\gamma\sigma^2 e}{2A\Delta k}}. \end{aligned}$$

These formulas admit a clear economic interpretation. The spread $\delta^b + \delta^a \approx 2/k + \Delta\omega$ decomposes into a static component $2/k$, corresponding to a risk-neutral monopolist facing exponential demand, and a dynamic risk premium reflecting inventory risk. The skew $\delta^b - \delta^a \approx 2q\omega$ is linear in inventory: a long market maker ($q > 0$) quotes more conservatively to reduce exposure.

The framework extends naturally to d correlated assets. In this case, optimal quotes depend on the inventory vector through a matrix

$$\Gamma = D^{-1/2}(D^{1/2}\Sigma D^{1/2})^{1/2}D^{-1/2},$$

where Σ is the variance-covariance matrix and D encodes the curvature of the Hamiltonian. The off-diagonal terms $\Gamma^{ij}q^j$ generate cross-asset effects: holding inventory in asset j modifies the optimal quotes for asset i through the correlation structure. In particular, when assets are positively correlated, a long position in one asset leads to more conservative quotes in the others.

3 Replication Methodology

3.1 Parameters and Conventions

We use the parameters from the paper's Section 6, summarised in Table 1. Inventory is represented in lots: $q = n\Delta$ for $n \in \{-Q, \dots, +Q\}$. We set $\ell \equiv 0$ throughout, studying the asymptotic regime where the terminal condition has negligible influence. All intensities are symmetric: $\Lambda^b = \Lambda^a = Ae^{-k\delta}$.

Parameter	IG	HY
σ ($\$/\sqrt{\text{s}}$)	5.83×10^{-6}	2.15×10^{-5}
A (s^{-1})	9.10×10^{-4}	1.06×10^{-3}
k ($\$^{-1}$)	1.79×10^{4}	5.47×10^{3}
Δ ($)	5×10^{7}	1×10^{7}
Q (lots)	4	4
γ ($\$^{-1}$)	6×10^{-5}	
ρ (IG–HY)	0.9	
T (s)	7200 (2 h)	

Table 1: Calibrated parameters from Guéant (2017).

3.2 Numerical Solvers

For Model A with exponential intensities, we exploit the v-transform $v_q = \exp(k\theta_q/\Delta)$, which linearises the nonlinear ODE into a tridiagonal system. This system is solved by implicit backward Euler, yielding a significantly faster scheme than generic nonlinear solvers.

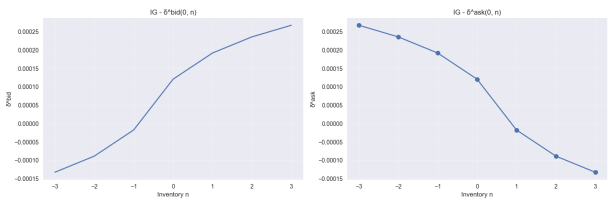
For general ξ (including Model B), we solve the nonlinear ODE directly using an implicit Euler scheme combined with Newton iteration, where both the residual and its tridiagonal Jacobian are evaluated at each time step.

In the multi-asset case, the problem is solved on the Cartesian grid $\{-Q_1, \dots, Q_1\} \times \{-Q_2, \dots, Q_2\}$ using the same implicit scheme, with a sparse Jacobian reflecting the higher-dimensional structure.

4 Replication Results

4.1 Single-Asset Quote Profiles

The quotes display two key properties.



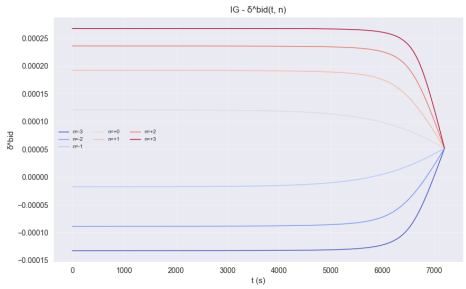


Figure 1: Top: optimal bid (left) and ask (right) distances versus inventory ($t = 0$, Model A, IG index). The bid increases monotonically with inventory; the ask decreases. Bottom: bid distance versus time for each inventory level.

First, the market maker skews quotes to mean-revert inventory: a long position ($n > 0$) widens the bid and tightens the ask to encourage selling. Second, the quotes reach their asymptotic regime well before the terminal time. At $t = 0$, they are indistinguishable from those at $t = T - 3600\,\text{s}$, justifying all subsequent evaluations in the stationary regime.

The spread is widest at zero inventory and decreases toward the boundaries $|n| \to Q$: at extreme positions, the market maker quotes aggressively through mid on one side (negative δ), compressing the total spread.

The skew vanishes at $n = 0$ by symmetry and increases nonlinearly : concave for $n > 0$, convex for $n < 0$. Both quantities deviate from the closed-form predictions (constant spread, linear skew), confirming that the exact ODE solution captures meaningful inventory-dependent curvature.

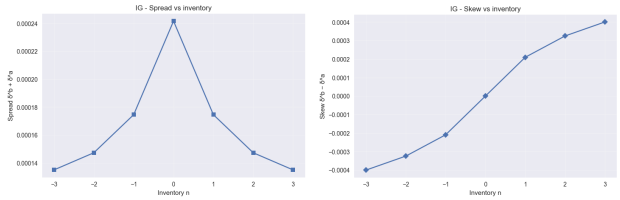


Figure 2: Spread $\delta^b + \delta^a$ (top) and skew $\delta^b - \delta^a$ (bottom) versus inventory for the IG index ($t = 0$, Model A).

4.2 Exact ODE versus Closed-Form Approximation

The GLF approximations provide operationally useful formulas but their accuracy depends on the parameter regime. Figure 3 overlays the exact ODE solution and the closed-form approximation for the IG index.

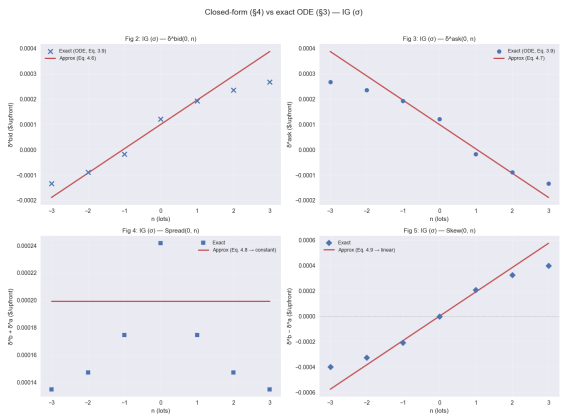


Figure 3: Exact ODE solution (markers) versus GLF closed-form approximation (lines) for the IG index at normal volatility. Top: bid and ask. Bottom: spread and skew. The approximation imposes a constant spread and linear skew, while the exact solution exhibits clear curvature.

The spread is largest at zero inventory and decreases toward the boundaries $|n| \to Q$. At extreme positions, the market maker quotes aggressively on one side (possibly through mid, i.e. negative δ), which compresses the total spread.

The skew vanishes at $n = 0$ by symmetry and increases nonlinearly: it is concave for $n > 0$ and convex for $n < 0$, indicating that the adjustment of quotes weakens near inventory limits.

Both features deviate from the GLF approximation, which imposes a constant spread $(2/k + \Delta\omega)$ and a linear skew $(2n\omega\Delta)$. This shows that the exact ODE solution captures non-trivial inventory-dependent curvature.

The approximation quality is governed by the ratio $\gamma\sigma^2/(A\Delta k C_\xi)$. When this ratio is small, the inventory dependence of θ is weak and the approximation is accurate. At lower volatility $(\sigma/2)$, the two solutions become nearly indistinguishable. In contrast, under higher volatility or lower liquidity, deviations increase significantly near $|n| = Q$.

This yields a clear practical implication: closed-form formulas are reliable in stable regimes, but the full ODE solution is required in stressed conditions.

4.3 Model A versus Model B

Under the paper's calibration, the quotes produced by Model A $(\xi = \gamma)$ and Model B $(\xi = 0)$ are numerically very close. Figure 4 shows the overlay for the IG index.

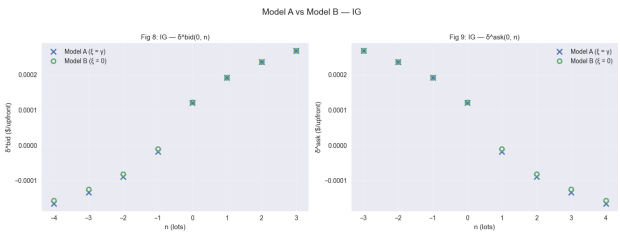


Figure 4: Model A $(\xi = \gamma)$ versus Model B $(\xi = 0)$ for the IG index. The two models produce nearly indistinguishable quotes under the calibrated parameters $(\gamma\Delta/k \approx 0.17)$.

The maximum absolute difference in bid distance across the inventory grid is less than 1% of the baseline spread. This closeness arises because $\gamma\Delta/k$ is small: in the IG case, $\gamma\Delta/k \approx 0.17$, so the exponential correction in the Model A Hamiltonian is close to the linear limit of Model B. This confirms the paper's practical conclusion that Model B (algebraically simpler and with a more transparent running-penalty interpretation) is a reliable substitute. The closeness is, however, parameter-dependent: at much higher risk aversion or with larger lot sizes, the two models can diverge.

4.4 Multi-Asset Cross-Inventory Effects

In the two-asset setting (IG and HY with $\rho = 0.9$), the ODE solver recovers the cross-inventory effects documented in the paper. Figure 5 displays the IG bid distance as a function of both inventories.

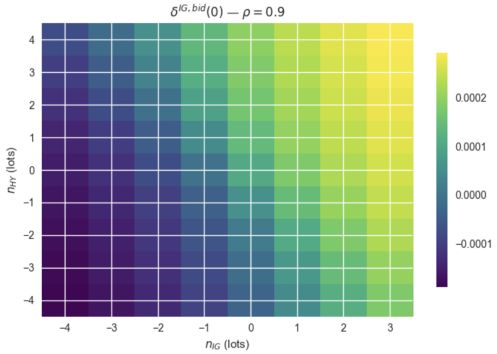


Figure 5: Multi-asset IG bid distance as a function of both IG and HY inventories $(\rho = 0.9,\ t = 0)$, reflecting cross-asset risk coupling. Warmer colours indicate wider (more conservative) quoting.

The key finding is that the optimal bid for asset i depends not only on inventory in asset i, but also on inventory in the correlated asset j. In particular, being long HY while quoting IG leads to a more conservative IG bid, although this cross-effect is weaker than the direct dependence on IG inventory.

This reflects the correlation structure: a long position in HY already exposes the market maker to similar risk, reducing the incentive to accumulate additional IG inventory. The strength of this cross-effect is controlled by ρ. At $\rho = 0$, the two assets decouple and the solution reduces to independent single-asset problems. At $\rho = 0.9$, the cross-effect becomes clearly visible, though smaller in magnitude than the direct inventory effect.

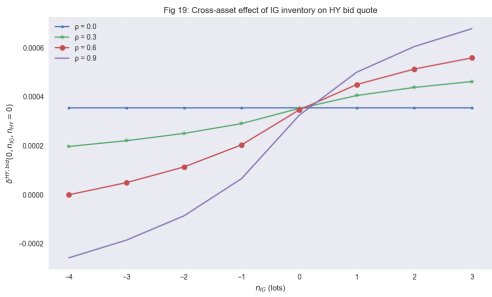


Figure 6: Cross-asset effect: HY bid distance versus IG inventory for several values of ρ (with $n_{\mathrm{HY}} = 0$). The effect vanishes at $\rho = 0$ and strengthens monotonically with correlation.

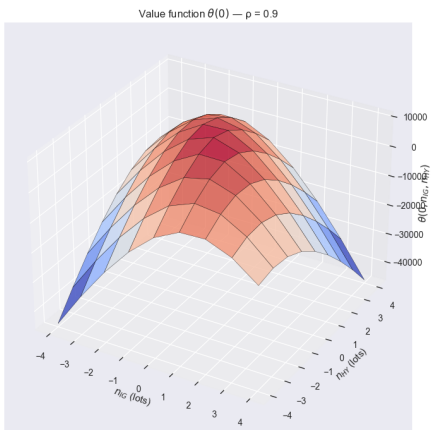


Figure 7: Value function $\theta(0, n_{\mathrm{IG}}, n_{\mathrm{HY}})$ at $\rho = 0.9$. The surface is concave, attaining its maximum at $(0, 0)$ and decaying toward the corners as the quadratic inventory cost $\frac{1}{2}\gamma\sigma^2 q^\top \Sigma q$ dominates.

The value function surface provides the geometric intuition behind the cross-inventory effects: θ is jointly maximised at zero inventory, so any deviation in either asset reduces the value and pushes quotes toward mean-reversion. The steeper descent along the IG axis reflects the larger lot size Δ_{IG}, which amplifies the per-lot risk contribution relative to HY.

This analysis confirms the practical value of the multi-asset extension: a market maker who ignores correlation and treats each asset independently would quote suboptimally in the presence of cross-holdings.

5 Validation and Numerical Extensions

5.1 Monte Carlo Validation

The ODE solution provides optimal quotes in theory and Monte Carlo simulation tests whether these quotes produce the predicted risk-adjusted performance in practice. We simulate 2 000 episodes using the IG parameters and compare the optimal strategy (ODE lookup table) against a naive benchmark that quotes a fixed symmetric spread equal to the optimal half-spread at zero inventory.

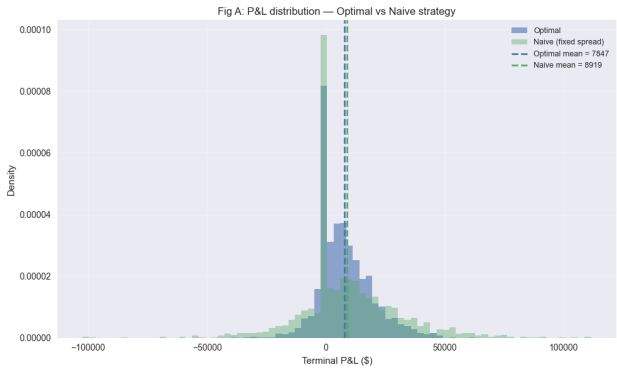


Figure 8: Terminal PnL distribution over 2 000 Monte Carlo paths: optimal (ODE) versus naive (fixed spread).

Figure 8 reveals a pattern that is easy to misunderstand. The naive strategy achieves a higher mean PnL ($8 919 vs. $7 847) despite generating fewer fills (1.5 vs. 2.7 per path on average). The optimal strategy fills more frequently because it actively tightens quotes on the mean-reverting side whenever inventory deviates from zero, attracting additional executions to rebalance the position. The naive strategy, quoting a fixed symmetric spread at all times, never exploits this mechanism. The cost of this conservatism is borne entirely in risk: the naive strategy runs at nearly twice the standard deviation ($20 784 vs. $11 081). The Sharpe ratio of the optimal strategy (0.71) exceeds that of the naive (0.43) by +65%, and the 5th percentile improves from −$20 023 to −$5 841, a fourfold reduction in worst-case losses.

This confirms a point that is essential for interpreting the Guéant framework: **Model A maximises CARA utility, *not* expected PnL**. The optimal strategy deliberately sacrifices mean revenue for variance reduction. A strategy that underperforms the naive benchmark on mean PnL while outperforming it on Sharpe ratio and certainty equivalent is behaving exactly as the theory predicts.

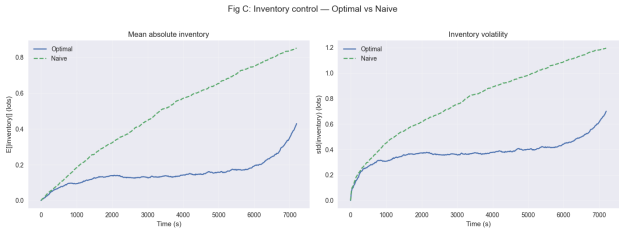


Figure 9: Mean absolute inventory over time for the optimal and naive strategies. The optimal strategy actively mean-reverts inventory toward zero; the naive strategy allows it to drift.

Figure 9 shows the mechanism behind this performance difference. The optimal strategy keeps inventory near zero throughout the horizon by continuously skewing quotes, while the naive strategy, which ignores inventory, allows positions to accumulate. This is the operational meaning of the ODE-derived policy: inventory management, not profit maximization.

5.2 Empirical Backtest on Real Order-Book Data

To test the framework beyond synthetic simulations, we calibrated the model to real limit-order-book data in two stages, starting with cryptocurrency data and then moving to US equities.

Crypto calibration (Tardis/Binance Futures). We first calibrated on Level-1 snapshots for BTCUSDT and ETHUSDT from the Tardis feed (Binance Futures), covering six daily sessions sampled monthly over October 2024 – March 2025. The volatility σ is estimated from intra-session mid-price returns. For the fill intensity (A, k), we use a hypothetical-quote maximum likelihood procedure: we construct a grid of quote distances δ from the observed mid-price, compute for each δ the passive exposure time and the number of intervals in which an aggressive trade would have crossed the hypothetical quote, and fit $\Lambda(\delta) = Ae^{-k\delta}$ by profile MLE with analytic solution $A^*(k) = \sum N_j / \sum E_j e^{-k\delta_j}$.

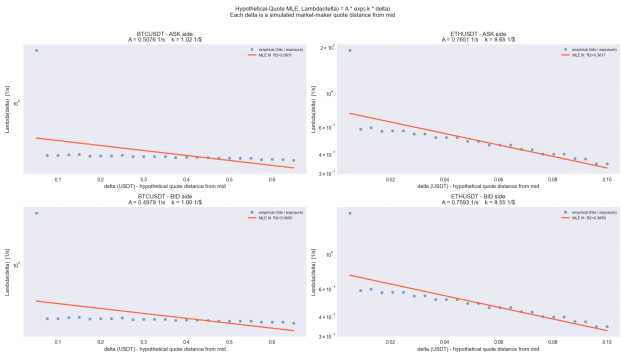


Figure 10: Hypothetical-quote MLE intensity fit for BTCUSDT and ETHUSDT (ask and bid sides). For BTC the fit is very poor ($R^2 \approx 0.06$), indicating that the exponential model is not well supported by L1 crypto data. ETH is marginally better ($R^2 \approx 0.36$) but still weak.

As Fig. 10 shows, the exponential model fails to fit the crypto intensity curves: the empirical $\hat{\Lambda}(\delta)$ is nearly flat across the delta grid, whereas $Ae^{-k\delta}$ decays rapidly. This reflects two structural features of the crypto L1 data. First, only top-of-book information is available: without queue

depth, the estimated A and k are reduced-form proxies rather than structural execution parameters. Second, BTC and ETH both exhibit very low decay rates ($k_{\text{BTC}} \approx 1.0$, $k_{\text{ETH}} \approx 8.6$) compared to the paper's CDX values ($k \approx 10^4$), meaning quote distance has little effect on execution probability, a symptom of high volatility relative to the spread. We therefore do not run a full backtest on crypto data and instead use the LOBSTER dataset for the empirical evaluation.

LOBSTER calibration (US equities). We calibrate on LOBSTER Level-10 order-book data for AAPL, SPY, and TSLA over three regular trading days in March 2026, filtered to regular trading hours (09:30–16:00 ET). The availability of ten levels of depth enables a queue-aware fill approximation: for each hypothetical quote at n ticks from the same-side best price, we approximate the displayed queue ahead using visible depth and classify an interval as a fill only if the aggressive volume reaching the quote price exceeds that queue. This is a strictly more conservative estimate than the touch proxy used for crypto. The resulting calibrated parameters are shown in Table 2.

	AAPL	SPY	TSLA
σ ($\$/\sqrt{\text{s}}$)	0.0259	0.0417	0.0549
A (s^{-1})	0.163	0.223	0.267
k ($\$^{-1}$)	83.0	93.9	45.0
Δ ($)	259.9	679.4	402.0
Median half-spread	$0.015	$0.005	$0.030

Table 2: Calibrated LOBSTER parameters (3 days, queue-aware fill approximation). SPY has the tightest spread (≈ 0.15 bps), consistent with its position as the most liquid US equity.

The large k values reflect the discrete tick structure of US equities: execution probability falls sharply with each additional tick of distance, consistent with a deep book concentrated near the best quotes. The ratio $\gamma\sigma^2/(A\Delta k)$ is small for all three assets, placing them in the regime where the GLF closed-form approximation should be accurate.

Backtest design. Using real mid-prices and a deterministic fill engine (a fill occurs when a real aggressive trade crosses the market maker's posted level, subject to: quote clamping to at least the current half-spread, tick discretisation to the $0.01 grid, ambiguous intervals skipped, and a terminal liquidation cost $|q_T|\Delta \cdot \text{hs}_T$), we run rolling 30-minute windows with 15-minute stride over each session. We compare four strategies: Optimal A (ODE, $\xi = \gamma$), Optimal B ($\xi = 0$), Closed-form (GLF), and Naive (symmetric median half-spread). The risk aversion γ is selected per asset by a preliminary Sharpe-ratio sweep on 20 windows.

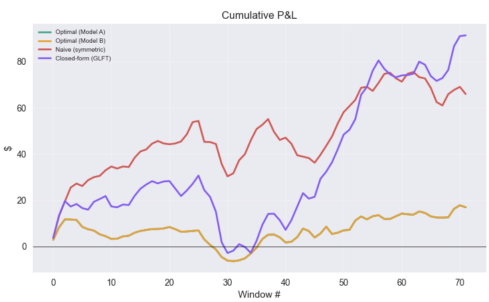


Figure 11: Backtest results for SPY (72 windows, $\gamma = 10^{-4}$, $\Delta = 1$ share).

Results and interpretation. The results are asset-dependent. For AAPL and TSLA, the optimal strategy outperforms the naive benchmark on Sharpe ratio, consistent with the Monte Carlo findings. For SPY, the naive and closed-form strategies achieve higher cumulative P&L, while Optimal A wins only $\sim 34\%$ of windows against the naive. Two factors explain this. First, SPY has a median half-spread of $0.005 (one tick), making the effective quote space nearly binary: any strategy that posts at the best price fills at the same rate. In this regime, inventory skewing via the ODE provides little additional value over a fixed symmetric quote. Second, Model A and Model B produce nearly identical quotes throughout (consistent with the small $\gamma\Delta/k$ ratio), confirming the near-equivalence result of Section 4.3 in a real data setting.

Two structural observations apply across all assets. First, parameter scales are physically incomparable to the paper's CDX indices ($k \sim 10^4$, $\Delta \sim \$50\text{M}$): the dimensionless quantity $\gamma\sigma^2/(A\Delta k)$ is the appropriate basis for comparison. Second, empirical fills exhibit clustering inconsistent with the Poisson model, providing direct motivation for the Hawkes extension.

5.3 Complementary ideas

General intensity functions. The ODE framework holds for any Λ satisfying conditions (H1)–(H4), not only the exponential form used in Section 3. We verified this with two additional families calibrated to the same $\Lambda(0) = A$ as the IG baseline: a power-law $\Lambda(\delta) = A(1 + k\delta)^{-3}$ and a logistic $\Lambda(\delta) = A/(1 + e^{k(\delta - \delta_0)})$. Both solvers converge and produce well-defined quotes.

The intensity shape enters the control problem only through two quantities evaluated at $p = 0$: the static offset $\delta^*(0)$, which sets the spread level, and the Hamiltonian curvature $H_0''(0)$, which determines the inventory-skew slope $\omega = \sqrt{\gamma\sigma^2/(2H_0''(0))}$. The power-law, with its fatter tails, yields a wider spread and shallower skew — fills remain probable at large distances so the market maker quotes less aggressively. The logistic, with its sharp cutoff near δ_0, produces a tighter spread and steeper skew. Across all families, the qualitative structure is preserved: the skew is approximately linear in inventory and the spread is approximately constant at first order. What differs are the quantitative coefficients $\delta^*(0)$ and ω, confirming the generality of Sections 3–4 beyond the exponential special case.

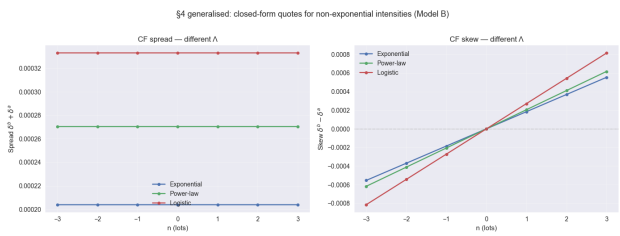


Figure 12: Closed-form spread (left) and skew (right) under exponential, power-law, and logistic intensities (IG parameters, Model B). The spread level and skew slope differ across families, but the qualitative inventory-dependent structure is preserved in all cases.

Terminal penalty sensitivity. The terminal condition $\theta(T, q) = -\ell(|q|)$ is numerically invisible at long horizons: for $T = 7200$ s, $|\theta(0, Q)|$ exceeds $\ell(Q\Delta)$ by several orders of magnitude, so the boundary condition is applied correctly

but drowned by the asymptotic solution at $t = 0$. The effect becomes visible at $T = 300\,\mathrm{s}$ with penalty coefficients $\alpha \geq 10$ (where $\ell(Q\Delta) = \alpha \cdot \mathrm{hs}$).

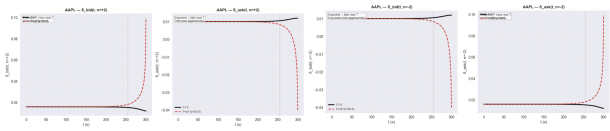


Figure 13: Asymmetric effect of the terminal penalty near maturity ($T = 300\,\mathrm{s}$) for AAPL ($\alpha = 20$). For a long inventory ($n = +2$), the ask quote becomes more aggressive (lower δ^a) while the bid becomes more conservative (higher δ^b); the opposite holds for $n = -2$. Results for SPY and TSLA are qualitatively identical and omitted for readability.

The penalty acts *asymmetrically* depending on inventory sign. A market maker with $n = +2$ lots near maturity lowers δ^a to sell aggressively and raises δ^b to avoid increasing the position; the mirror pattern holds for $n = -2$. The total spread $\delta^b + \delta^a$ is therefore the wrong diagnostic: it can increase, decrease, or remain flat depending on which effect dominates. The economically correct description is an inventory-dependent distortion of the quoting policy toward terminal flattening, monotone in the penalty strength α.

Three-asset extension. The d-asset framework of Section 5 extends beyond $d = 2$ without structural change. We built a general sparse Newton solver on the Cartesian inventory grid $(2Q+1)^d$ and tested it with $d = 3$: IG and HY as before (with $Q = 2$) plus a synthetic third asset, with correlations $\rho_{12} = 0.9$, $\rho_{13} = 0.7$, $\rho_{23} = 0.6$.

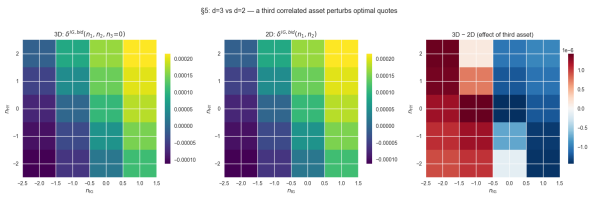


Figure 14: Impact of a third correlated asset on optimal IG bid quotes. Left: 3-asset model ($d = 3$) with $n_3 = 0$. Middle: corresponding 2-asset model. Right: difference between the two, highlighting the marginal effect of the third asset.

Figure 14 compares the IG bid quote surface obtained in the 3-asset model to the 2-asset benchmark. The difference highlights the marginal impact of the third asset through the covariance structure. While the effect remains modest under this calibration, it is systematic and reflects the extension of cross-inventory coupling to higher dimensions. The small magnitude of the difference suggests that higher-order interactions remain second-order effects under realistic parameter values.

6 Extension: Hawkes Execution Processes

We develop our extension in the single-asset setting and under the Model B objective ($\xi = 0$), which is the most widely used formulation in the recent literature on algorithmic market making [4]. As shown in Section 3, the differences between Model A and Model B are small in practice (see also Figures 8–9 and 15–16 in [1]), and the single-asset case suffices to demonstrate the structural consequences of replacing Poisson arrivals with Hawkes processes. Furthermore, we restrict to pure self-excitation ($\alpha_{ab} = \alpha_{ba} = 0$), so that a bid fill increases future bid intensity but has no effect on ask intensity, and conversely. This isolates the effect of temporal clustering on each side of the book independently, and the extension to cross-excitation is straightforward within the same framework.

6.1 Motivation: Limitations of Poisson Arrivals

The Poisson execution assumption underlying implies three properties that are empirically counterfactual:

(i) *Absence of clustering*: Poisson processes have independent increments and exponentially distributed inter-arrival times. Real order flow exhibits short-term clustering and overdispersion: the occurrence of a trade increases the probability of subsequent trades.

(ii) *No feedback from past executions*: In the baseline model, the intensity at time t depends only on the current quote distance. Past trades have no effect on future execution rates.

(iii) *No cross-excitation*: Even in the multi-asset extension, intensities remain independent across instruments. In practice, trades in correlated assets frequently generate spillover effects.

Figure 15 illustrates the self-exciting mechanism. Each fill (vertical tick) causes a jump in the excitation process $y(t)$, which temporarily raises the intensity $\lambda(t)$ above its baseline level. When fills occur in rapid succession (around $t \approx 1\,\mathrm{s}$), the excitation accumulates, producing bursts of high intensity followed by quiet periods as y decays back to zero. This clustering pattern is absent from Poisson processes, where the intensity is constant between control updates.

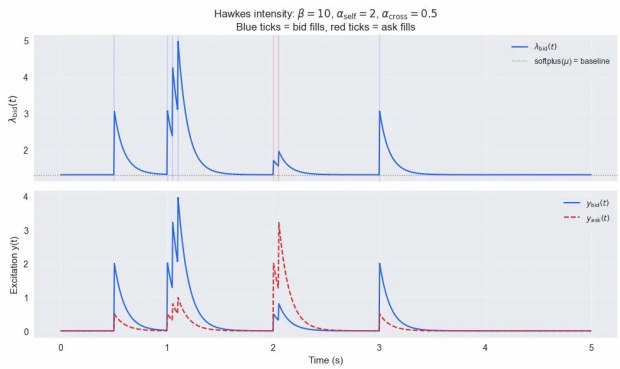


Figure 15: Hawkes excitation dynamics. Top: bid intensity $\lambda_{\mathrm{bid}}(t)$ with fill events (vertical ticks). Bottom: excitation processes $y_{\mathrm{bid}}(t)$ and $y_{\mathrm{ask}}(t)$. Parameters: $\beta = 10$, $\alpha_{\mathrm{self}} = 2$, $\alpha_{\mathrm{cross}} = 0.5$.

Motivated by these observations, we extend the model by replacing Poisson arrivals with Hawkes processes, following the additive specification introduced in Gašperov and Kostanjčar [4].

6.2 Modified Arrival Intensities

We replace the Poisson intensities with:

$$\lambda_t^b = Ae^{-k\delta_t^b} + y_t^b, \qquad \lambda_t^a = Ae^{-k\delta_t^a} + y_t^a,$$

where $y_t^b, y_t^a \geq 0$ are excitation processes satisfying:

$$dy_t^b = -\beta\, y_t^b\, dt + \alpha_{bb}\, dN_t^b + \alpha_{ba}\, dN_t^a,$$
$$dy_t^a = -\beta\, y_t^a\, dt + \alpha_{aa}\, dN_t^a + \alpha_{ab}\, dN_t^b.$$

Here $\beta > 0$ is the decay rate of excitation (so that $1/\beta$ is the memory time scale), and $\alpha_{ij} \geq 0$ are excitation coefficients. Between jumps, y^b decays exponentially toward zero; at each bid fill, y^b jumps by α_{bb} (self-excitation) and y^a jumps by α_{ab} (cross-excitation).

The process is stationary if and only if the spectral radius of the matrix $K = (\alpha_{ij}/\beta)_{i,j}$ is strictly less than one. In our experiments, $\alpha_{bb} = \alpha_{aa} = 0.3$, $\alpha_{ab} = \alpha_{ba} = 0$, and $\beta = 1.5$, giving a spectral radius of 0.2.

6.3 Markov Property of the Augmented State

The counting process N_t^b alone is non-Markovian: its intensity depends on the entire history of past arrivals through the integral $\int_0^t \alpha\, e^{-\beta(t-s)}\, dN_s^b$. However, the exponential kernel gives a crucial structural property.

Lemma 6.1 (Markovianity of the augmented process)**.** *Consider a Hawkes process with exponential kernel satisfying $dy_t = -\beta\, y_t\, dt + \alpha\, dN_t$. Then the pair (N_t, y_t) is a Markov process.*

Proof. Between consecutive jump times, y_t satisfies the deterministic ODE $\dot{y}_t = -\beta\, y_t$, with explicit solution $y_t = y_s\, e^{-\beta(t-s)}$ for $t \in (s, \tau_{n+1})$, where τ_{n+1} is the next jump time after s. At each jump time τ, the excitation increments instantaneously: $y_\tau = y_{\tau^-} + \alpha$. In both regimes, the future evolution of y_t depends only on its current value—not on the past trajectory of N. Since $\lambda_t = Ae^{-k\delta_t} + y_t$ is a function of y_t and the current control, the conditional distribution of future increments of N_t depends only on (N_t, y_t). □

This result is what allows us to write a HJB equation for the extended problem. The Markov state of the control problem is:

$$Z_t = (t,\, S_t,\, X_t,\, q_t,\, y_t^b,\, y_t^a).$$

which has six components, compared to four (t, S_t, X_t, q_t) in the Poisson case. After the CARA ansatz eliminates (S_t, X_t), the reduced state is (t, q, y^b, y^a), four dimensions instead of two. It is this increase in dimensionality that prevents the ODE reduction.

6.4 HJB Equation and Failure of Dimensional Reduction

Applying the linear ansatz $u = X + qS + \theta(t, q, y^b, y^a)$ from Model B eliminates the dependence on (S, X) as before. The reduced value function $\theta(t, q, y^b, y^a)$ satisfies:

$$\begin{aligned} 0 = &-\partial_t\theta + \tfrac{1}{2}\gamma\sigma^2 q^2 - \beta y^b \partial_{y^b}\theta - \beta y^a \partial_{y^a}\theta \\ &- \sup_{\delta^b}(Ae^{-k\delta^b} + y^b)\Big(\Delta\delta^b + \Theta^b\Big) \\ &- \sup_{\delta^a}(Ae^{-k\delta^a} + y^a)(\Delta\delta^a + \Theta^a)\,, \end{aligned}$$

with terminal condition $\theta(T, q, y^b, y^a) = -\ell(|q|)$ and where:

$$\Theta^b = \theta(t, q{+}1, y^b{+}\alpha_{bb}, y^a{+}\alpha_{ab}) - \theta(t, q, y^b, y^a),$$
$$\Theta^a = \theta(t, q{-}1, y^b{+}\alpha_{ba}, y^a{+}\alpha_{aa}) - \theta(t, q, y^b, y^a).$$

This equation differs from the preivous in three fundamental ways:

1. The drift terms $-\beta y^b \partial_{y^b}\theta$ and $-\beta y^a \partial_{y^a}\theta$ couple θ across the continuous dimensions (y^b, y^a).
2. The jump terms evaluate θ at shifted excitation levels $(y^b + \alpha_{bb}, y^a + \alpha_{ab})$, creating a nonlocal dependence.
3. The intensity prefactor $Ae^{-k\delta^b} + y^b$ mixes the control δ^b with the state variable y^b, so the first-order optimality condition for δ^{b*} has no closed-form solution when $y^b > 0$.

In particular, the ODE reduction that is the hallmark of the Guéant framework is *destroyed*: θ can no longer be written as a function of (t, q) alone.

Remark 6.2 (On the multiplicative specification)**.** One may ask whether the multiplicative form $\lambda_t^b = Ae^{-k\delta_t^b}(1 + y_t^b)$ could restore tractability, since the control δ^b enters only through the factor $Ae^{-k\delta^b}$. However, the jump increments $\Theta^b = \theta(t, q{+}1, y^b{+}\alpha, y^a{+}\alpha') - \theta(t, q, y^b, y^a)$ still couple inventory shifts with excitation shifts, preventing any factorisation of θ into a product $\phi(t, q) \cdot Z(y^b, y^a)$. We retain the additive specification consistent with [4].

6.5 Reinforcement Learning Approach

Since PDE cannot be solved analytically or reduced to an ODE system, we approximate the optimal policy numerically using deep reinforcement learning. Following [4], we train a Soft Actor–Critic (SAC) agent to learn the mapping

$$(t,\, q,\, y^b,\, y^a) \longmapsto (\delta^{b*},\, \delta^{a*}).$$

SAC is an off-policy actor-critic algorithm that maximises the expected cumulative reward augmented by an entropy bonus, where the entropy coefficient is automatically tuned. The entropy regularisation encourages exploration and prevents premature convergence, which is valuable in our stochastic environment where the reward signal is noisy (due to the Brownian price component).

The environment is implemented as a Gymnasium environment. At each time step Δt, the agent observes the normalised state vector $(t/T,\, q/Q,\, y^b/y_{\max},\, y^a/y_{\max})$ and outputs an action in $[-1, 1]^2$, which is linearly mapped to quote distances $(\delta^b, \delta^a) \in [\delta_{\min}, \delta_{\max}]$. Inventory constraints are enforced by setting $\delta^b = \delta_{\max}$ when $q = Q$ (suppressing buys) and $\delta^a = \delta_{\max}$ when $q = -Q$ (suppressing sells). The Hawkes simulator then samples fills, updates the excitation variables, and evolves the price.

The reward at each step is the incremental Model B objective:

$$r_t = \big(\mathrm{PnL}_{t+1} - \mathrm{PnL}_t\big) - \tfrac{1}{2}\gamma\sigma^2 q_t^2\, \Delta t,$$

with an additional terminal penalty $-\ell_{\mathrm{liq}}\, q_T^2$ at the final step. This decomposition ensures that the sum of rewards over an episode equals the Model B objective up to the terminal penalty term, which serves as a practical incentive for the agent to reduce inventory near the end of the horizon.

6.6 Experimental Setup

We simulate a single-asset market making environment with Hawkes order arrivals as described in Section 4. All parameters are synthetic and chosen to produce a realistic balance between spread revenue, inventory risk, and self-excitation effects.

Param.	Value	Rationale
T	20 s	Short horizon for training
Δt	0.1 s	$< 1/\beta = 0.67$ s
σ	0.05	$\sigma\sqrt{T}$=0.22≈$1/k$
γ	0.5	Moderate risk aversion
A	1.5	Fill rate Ae^{-1}=0.55/s
k	1.5	Distance scale $1/k$=0.67
Q	5	Inventory limits ±5
ℓ_{liq}	0.5	ℓQ^2=12.5≈ PnL
α_{bb}=α_{aa}	0.3	α/β=0.2 < 1
α_{ab}=α_{ba}	0.0	No cross-excitation
β	1.5	Memory $1/\beta$=0.67 s

Table 3: Simulation parameters.

Each episode consists of $T/\Delta t = 200$ steps. The SAC agent uses a two-layer neural network (128 units per layer) with automatic entropy tuning, a replay buffer of 500,000 transitions, batch size 512, and a linearly decaying learning rate from 3×10^{-4} to 10^{-5} over 800,000 training steps. The discount factor is $\gamma_{\text{RL}} = 1$ (no discounting), consistent with the finite-horizon nature of the problem.

We compare the SAC agent against two Poisson-optimal baselines, both blind to the Hawkes excitation (y^b, y^a):

1. **GLFT-Exact**: the exact solution of the Guéant ODE system for Model B ($\xi = 0$) with $\ell = 0$, computed by implicit Euler with Newton iterations and evaluated at $t = 0$ (asymptotic regime). The optimal quotes are pre-computed as a lookup table indexed by inventory q.
2. **GLFT-Approx**: the closed-form asymptotic approximation.

Both baselines apply the optimal Poisson strategy in an environment where arrivals are in fact Hawkes. The comparison therefore isolates the value of observing and reacting to the excitation state.

6.7 Backtest Results

We evaluate each strategy over 500 independent episodes with identical seeds; the adjusted PnL is defined as $\text{PnL}_T - \ell_{\text{liq}}\, q_T^2$.

Strategy	PnL	$\lvert q\rvert_T$	PnL adj
SAC	17.21	1.18	15.96
GLFT-Exact	17.24	1.93	14.46
GLFT-Approx	16.61	2.44	12.41

Table 4: Backtest results over 500 episodes at $\alpha = 0.3$.

The SAC agent achieves the highest adjusted PnL (15.96), exceeding GLFT-Exact by +1.50 and GLFT-Approx by +3.55. The gain comes entirely from inventory management: raw PnL is nearly identical for SAC and GLFT-Exact (≈ 17.2), but terminal inventories differ sharply (1.18 vs. 1.93 vs. 2.44 lots). The ranking is stable across three disjoint seed ranges of 500 episodes each (Table 5), with the SAC advantage over GLFT-Exact consistently around +1.2.

Seed range	SAC	Exact	Approx
0–500	**15.79**	14.56	12.53
500–1000	**16.08**	14.67	12.67
1000–1500	**15.81**	14.73	13.15

Table 5: Robustness across seed ranges (500 episodes each).

Figure 16 illustrates the mechanism on a single episode. GLFT-Exact maintains a nearly constant spread of ≈ 1.4 throughout, while the SAC agent varies its spread between 0.8 and 2.8 — widening during excitation bursts (when the additive y component makes fills likely regardless of quote distance) and narrowing during quiet periods to attract profitable trades.

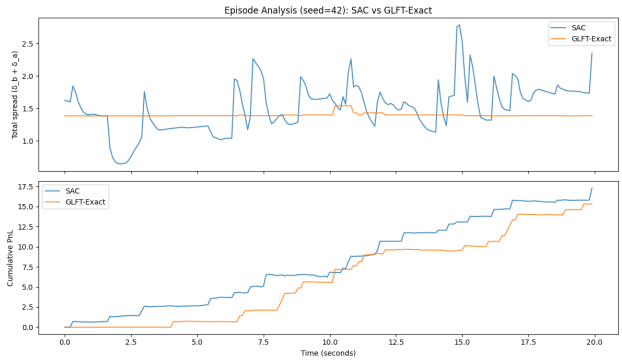


Figure 16: Single episode (seed 42): SAC vs GLFT-Exact. Top: total spread. Bottom: cumulative PnL.

6.8 Sensitivity to the Excitation Parameter

To isolate the contribution of Hawkes dynamics to the SAC advantage, we evaluate the same trained model (trained at $\alpha = 0.3$) across environments with different excitation levels, each over 200 episodes.

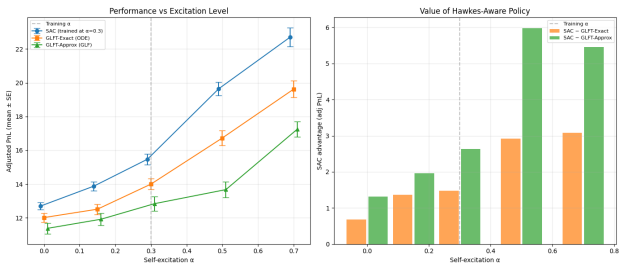


Figure 17: Left: adjusted PnL vs α. Right: SAC advantage over both baselines. The advantage increases monotonically with α.

α	SAC	Exact	SAC−Exact
0.00	12.69	12.02	+0.68
0.15	13.88	12.51	+1.37
0.30	15.47	14.00	+1.47
0.50	19.65	16.73	+2.92
0.70	22.70	19.62	+3.08

Table 6: Sensitivity analysis: adjusted PnL and SAC advantage as a function of α.

The results in Table 6 and Figure 17 highlight two observations.

First, the SAC advantage over GLFT-Exact increases monotonically with α: from +0.68 at $\alpha = 0$ (pure Poisson) to +3.08 at $\alpha = 0.7$ (strong excitation). This pattern highlights the central finding of the experiment: the value of a Hawkes-aware policy grows with the intensity of self-excitation, indicating that the agent genuinely exploits temporal clustering in order flow rather than merely learn-

ing a better static policy.

Second, the residual advantage at $\alpha = 0$ (+0.68) reflects the fact that the SAC agent is trained with a terminal penalty $\ell_{\text{liq}} q_T^2$ that the GLFT-Exact baseline does not incorporate in its ODE (which uses $\ell = 0$). The *incremental* advantage attributable to Hawkes awareness is therefore approximately $3.08 - 0.68 = 2.40$ at $\alpha = 0.7$, representing a +15% improvement over the Poisson-optimal strategy.

6.9 Discussion and Limitations

Strengths. The sensitivity analysis provides a clean identification of the Hawkes effect: the monotonic increase of the SAC advantage with α cannot be explained by the terminal penalty asymmetry, which is constant across α values. The comparison against both the exact ODE solution and the closed-form approximation provides a complete picture of the sources of improvement.

Limitations. Several limitations remain. First, all parameters are synthetic and not calibrated to a specific market. While the parameter choices ensure realistic scale relationships (Table 3), the absolute PnL figures should not be interpreted as predictions for any real trading environment. Second, the SAC agent is trained and evaluated in the same simulated environment with known Hawkes parameters (α, β). In practice, these parameters would need to be estimated online, introducing model uncertainty that could reduce the agent's advantage. Third, the terminal penalty asymmetry between the SAC training objective ($\ell_{\text{liq}} = 0.5$) and the ODE baseline ($\ell = 0$) means that the absolute advantage figures slightly overstate the pure Hawkes contribution; the incremental analysis above addresses this.

Possible improvements. Several directions could strengthen the results. Training the SAC agent at multiple values of α (curriculum learning) would test robustness to parameter uncertainty. Calibrating parameters to real order flow data. Incorporating cross-excitation ($\alpha_{ab} > 0$) would capture the empirically documented spillover effects between bid and ask sides. Finally, extending to multiple correlated assets with Hawkes arrivals would combine the multi-asset framework with the self-exciting dynamics.

An alternative hybrid approach, inspired by Falces Marin et al. (2022), would let the RL agent learn a state-dependent risk aversion $\gamma(t, q, y^b, y^a)$ while computing quotes analytically via the GLF formulas. This would combine the structural guarantees of the Guéant framework with the adaptivity of reinforcement learning, though it would restrict the policy to the family of GLF-optimal quotes parametrised by γ.

7 Conclusion

This report has replicated the core numerical experiments of Guéant (2017), validated the optimal strategy by Monte Carlo simulation, backtested it on real order-book data, and identified the precise point where the framework's tractability fails under more realistic dynamics.

All qualitative patterns of the paper are confirmed: inventory-dependent quote adjustment, rapid asymptotic convergence, U-shaped spread, nonlinear skew, Model A/B near-equivalence, and multi-asset cross-inventory coupling. The Monte Carlo validation shows that the optimal strategy's main advantage is risk control (Sharpe 0.71 versus 0.43), not mean PnL. It fills more frequently than the naive benchmark by actively tightening quotes on the mean-reverting side, deliberately trading revenue for variance reduction. The empirical backtest confirms inventory-management capability on AAPL and TSLA, while revealing that on SPY, where the half-spread is one tick, inventory-based quote adjustments become ineffective, highlighting the absence of queue dynamics in the model.

Replacing Poisson arrivals with additive Hawkes processes destroys the dimensional reduction: the reduced value function $\theta(t, q, y^b, y^a)$ satisfies a four-dimensional PDE with no ODE reduction. A SAC agent trained on this extended problem outperforms the Poisson-optimal baseline by an amount growing monotonically with self-excitation intensity, from +0.68 at $\alpha = 0$ to +3.08 at $\alpha = 0.7$, confirming that temporal clustering provides exploitable information the Guéant framework cannot capture. The course's emphasis on the branching ratio $\|\varphi\|_1$ is directly relevant: our environment operates at $\alpha/\beta = 0.2$, while empirical markets are often close to criticality, implying a potentially much stronger effect in practice.

The framework's main strengths, namely exact dimensional reduction, millisecond-computable quotes, unified ξ-parametrisation, and analytically tractable multi-asset extension, are all tightly coupled to the Poisson assumption. The mid-price also evolves independently of order flow, leaving adverse selection entirely unmodelled. The Guéant model should therefore be viewed not as a realistic description of modern electronic markets, but as a foundational baseline upon which more realistic, data-driven, and adaptive models must be built.

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